

FUNDAMENTALS OF LINUX OS

CEIS106 Final Course Project

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Introduction

How to utilize the Linux file system tree with Linux commands and execute shell scripts by experimenting with the standard input, user-defined variables, redirection, file permissions, and environment variables. Create, modify, and delete user group accounts using both CLI and GUI. Network IP configurations, system performance monitoring.

Tools used in this project - Google, Linux+ LPIC-1 Guide to Linux Certification, and Cengage.

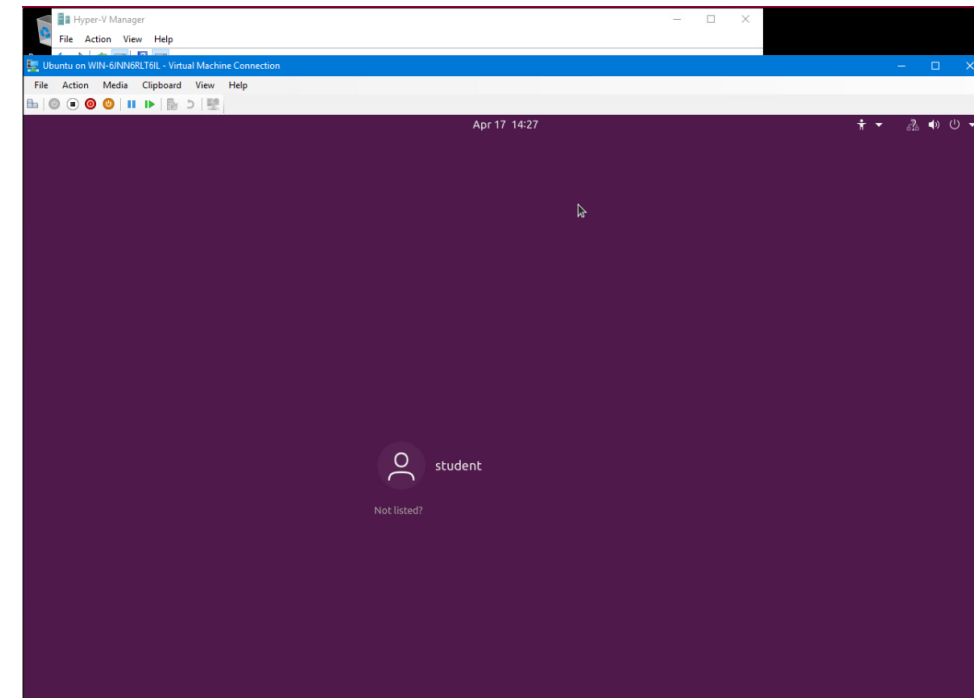
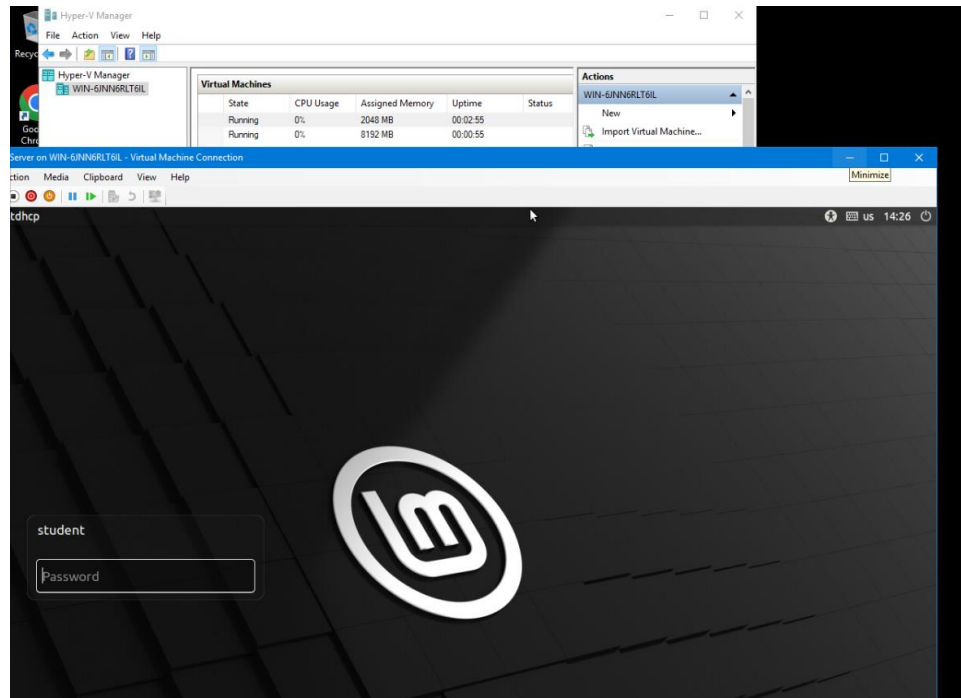
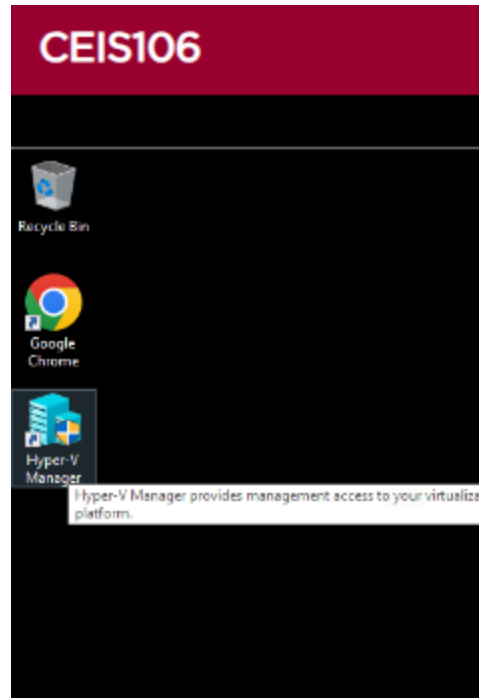
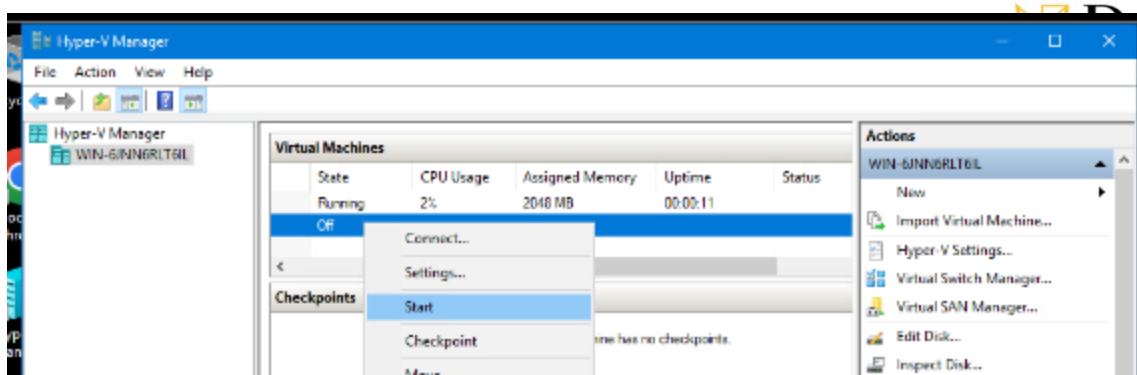
Course Project

Operating Systems

Module 2 Linux Filesystem Hierarchy

Exploring the Linux filesystem tree and common Linux commands used to manage directories and files.

Process of Starting Up Servers



Navigate the Linux filesystem tree

1. What is the pwd command an acronym for? What about the cd command?

Answer here: The pwd command is an acronym for print working directory for the present working directory. The cd command is for changing directories. When changing directories, you always want to verify your pwd, this is why we see pwd and cd.

2. Explain the differences between a relative path and an absolute/full path in Linux.

Answer here:

Absolute – From Root to a certain file/directory.

Relative – Current directory to a certain file/directory.

References:

1. Week 2 Team Live Session

2. Google

Create directories and files

```
student@ubuntu1: ~/JanFebSession/Course1
student@ubuntu1:~/JanFebSession/Course1$ tree -d -L 2 ~
/home/student
├── Desktop
│   └── linux-vm-tools
├── Documents
├── Downloads
├── JanFebSession
│   ├── Course1
│   ├── Course2
│   └── Course3
├── Music
├── Pictures
├── Public
├── shared-drives
├── snap
│   └── snap-store
├── Templates
└── Videos

16 directories
student@ubuntu1:~/JanFebSession/Course1$ ls -l ~/JanFebSession/Course1
total 0
-rw-rw-r-- 1 student student 0 Mar  7 11:32 file1
-rw-rw-r-- 1 student student 0 Mar  7 11:33 file2
-rw-rw-r-- 1 student student 0 Mar  7 11:33 file3
student@ubuntu1:~/JanFebSession/Course1$ echo Barbara Espericueta
Barbara Espericueta
student@ubuntu1:~/JanFebSession/Course1$ date
Thu 07 Mar 2024 11:52:38 AM EST
student@ubuntu1:~/JanFebSession/Course1$
```

Copy and remove directories and files

```
student@ubuntu1:~$ tree JanFebSession MarAprSession
JanFebSession
├── Course1
│   ├── file1
│   ├── file2
│   └── file3
├── Course2
└── Course3
MarAprSession
├── Course1
│   ├── file1
│   └── file2
└── Course2

5 directories, 5 files
student@ubuntu1:~$ echo Barbara Espericueta
Barbara Espericueta
student@ubuntu1:~$ date
Thu 07 Mar 2024 12:16:03 PM EST
student@ubuntu1:~$
```

Locate directories and files

```
student@ubuntu1: ~  
student@ubuntu1:~$ locate -r /file1$  
/home/student/JanFebSession/Course1/file1  
/home/student/MarAprSession/Course1/file1  
student@ubuntu1:~$ echo Barbara Espericueta  
Barbara Espericueta  
student@ubuntu1:~$ date  
Thu 07 Mar 2024 01:22:42 PM EST  
student@ubuntu1:~$
```


Course Project

Operating Systems

Module 3 Linux Shell Scripts

Creating and executing a Linux shell script by experimenting with the standard input, user-defined variables, redirection, file permissions, and environment variables.

Create a shell script

1. What are the file permissions of the script?

Answer here:

`-rw-rw-r—`

User+>rw(6), group=>rw(6), others=>r(4)

2. What's the name of the user-defined variable in the script?

Answer here:

text

3. Which redirection meta-character is used in the script? What does it do?

Answer here:

'>>', it redirects output to the file and append the data.

References:

1.Cengage

2.Live Class

Change script file permissions

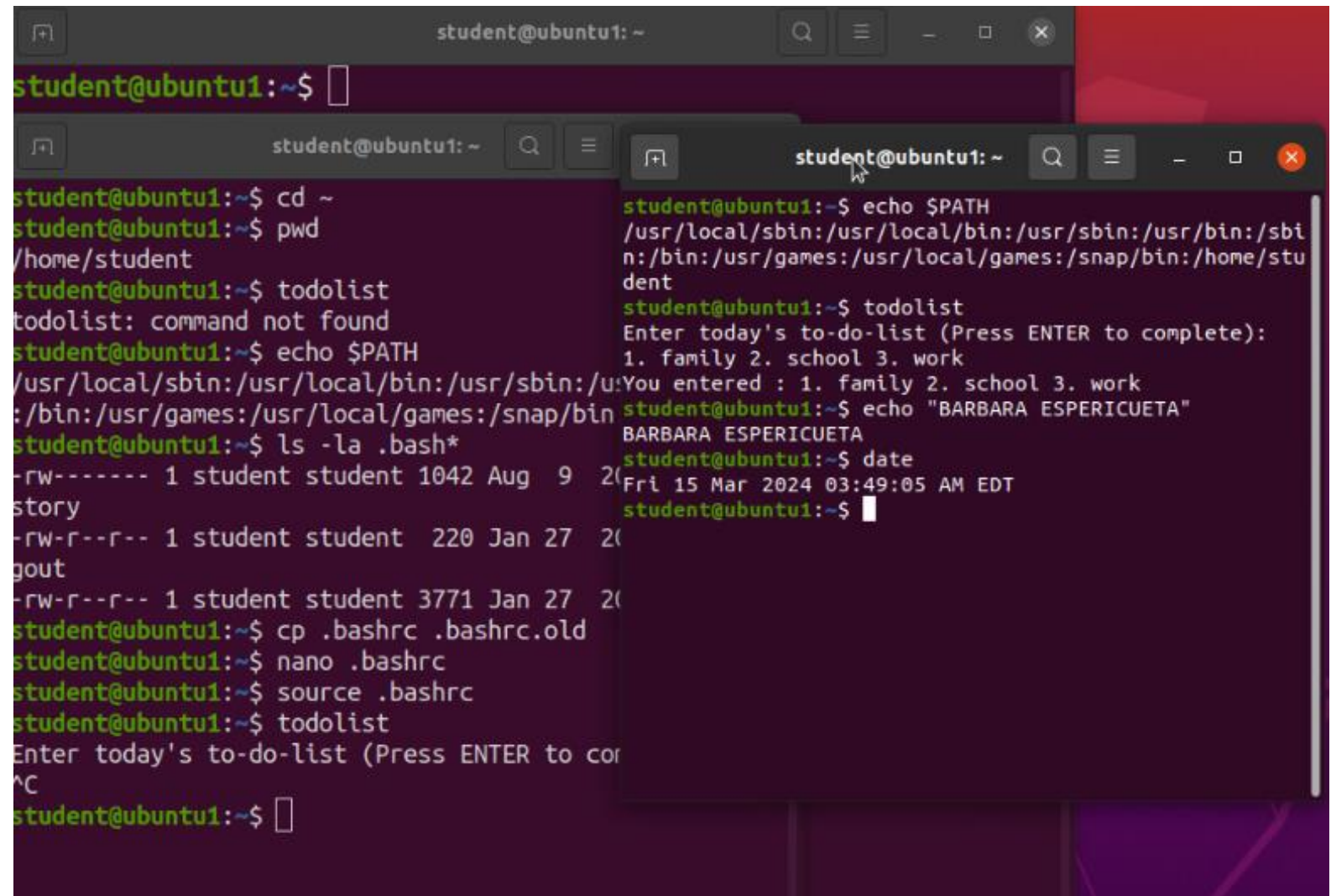
```
student@ubuntu1:~$ nano todolist
student@ubuntu1:~$ cat todolist
#!/bin/bash
echo "Enter today's to-do-list (Press ENTER to complete):"
read text
echo "You entered : $text"
date >> MyToDoList
echo "Today's to-do-list -- $text" >> MyToDoList
echo "" >> MyToDoList
student@ubuntu1:~$ ls -ltr todolist
-rw-rw-r-- 1 student student 200 Mar 15 03:12 todolist
student@ubuntu1:~$ chmod 755 todolist
student@ubuntu1:~$ ls -ltr todolist
-rwxr-xr-x 1 student student 200 Mar 15 03:12 todolist
student@ubuntu1:~$ ./todolist
Enter today's to-do-list (Press ENTER to complete):
1. work 2. family 3. school
You entered : 1. work 2. family 3. school
student@ubuntu1:~$ echo "BARBARA ESPERICUETA"
```

Set the PATH variable

```
student@ubuntu1:~$ cd ~
student@ubuntu1:~$ pwd
/home/student
student@ubuntu1:~$ todolist
todolist: command not found
student@ubuntu1:~$ echo $PATH
/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/usr/games:/usr/local/games:/snap/bin
student@ubuntu1:~$ PATH=$PATH:/home/student
student@ubuntu1:~$ echo $PATH
/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/usr/games:/usr/local/games:/snap/bin:/home/student
student@ubuntu1:~$ todolist
Enter today's to-do-list (Press ENTER to complete):
1. school 2. work 3. family
You entered : 1. school 2. work 3. family
student@ubuntu1:~$ echo "BARBARA ESPERICUETA"
BARBARA ESPERICUETA
student@ubuntu1:~$ date
Fri 15 Mar 2024 03:23:55 AM EDT
```


Make the PATH variable permanent

Run the *todolist* script before and after making the PATH variable permanent.



```
student@ubuntu1: ~$  
student@ubuntu1:~$ cd ~  
student@ubuntu1:~$ pwd  
/home/student  
student@ubuntu1:~$ todolist  
todolist: command not found  
student@ubuntu1:~$ echo $PATH  
/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/usr/games:/usr/local/games:/snap/bin  
student@ubuntu1:~$ ls -la .bash*  
-rw-r--r-- 1 student student 1042 Aug  9 2023 .bash_logout  
-rw-r--r-- 1 student student 220 Jan 27 2023 .bash_profile  
-rw-r--r-- 1 student student 3771 Jan 27 2023 .bashrc  
student@ubuntu1:~$ cp .bashrc .bashrc.old  
student@ubuntu1:~$ nano .bashrc  
student@ubuntu1:~$ source .bashrc  
student@ubuntu1:~$ todolist  
Enter today's to-do-list (Press ENTER to complete):  
^C  
student@ubuntu1:~$  
student@ubuntu1:~$ echo $PATH  
/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/usr/games:/usr/local/games:/snap/bin:/home/student  
student@ubuntu1:~$ todolist  
Enter today's to-do-list (Press ENTER to complete):  
1. family 2. school 3. work  
You entered : 1. family 2. school 3. work  
student@ubuntu1:~$ echo "BARBARA ESPERICUETA"  
BARBARA ESPERICUETA  
student@ubuntu1:~$ date  
Fri 15 Mar 2024 03:49:05 AM EDT  
student@ubuntu1:~$
```

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Module 4 User and Group Management

This module is to create, modify, and delete user and group accounts using both CLI and GUI.

Add users and groups in CLI

1. What does the `-m` option in the `useradd` command do?

Answer here:

`-m` -- Tells `useradd` command to create a home directory for the new user.

2. What does the `-3` option in the `tail` command do?

Answer here:

`-3` -- Tells `tail` command how many lines to show from the end of the file.

3. Which line of the `/etc/group` file lists members of the “students” group? Copy it here.

Answer here:

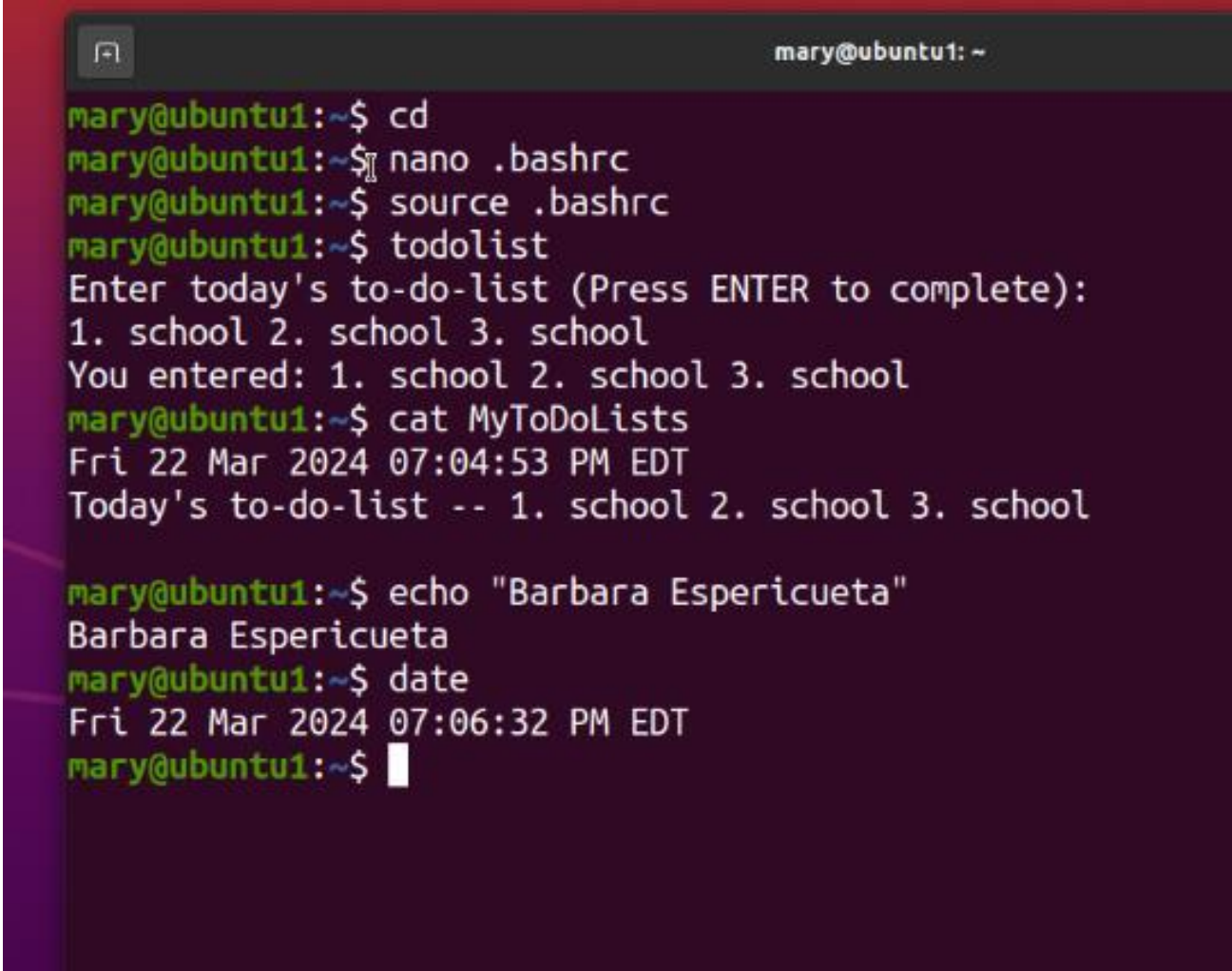


```
students:x:1002:student,mary
```

References:

1. Project Video

Test user and group settings



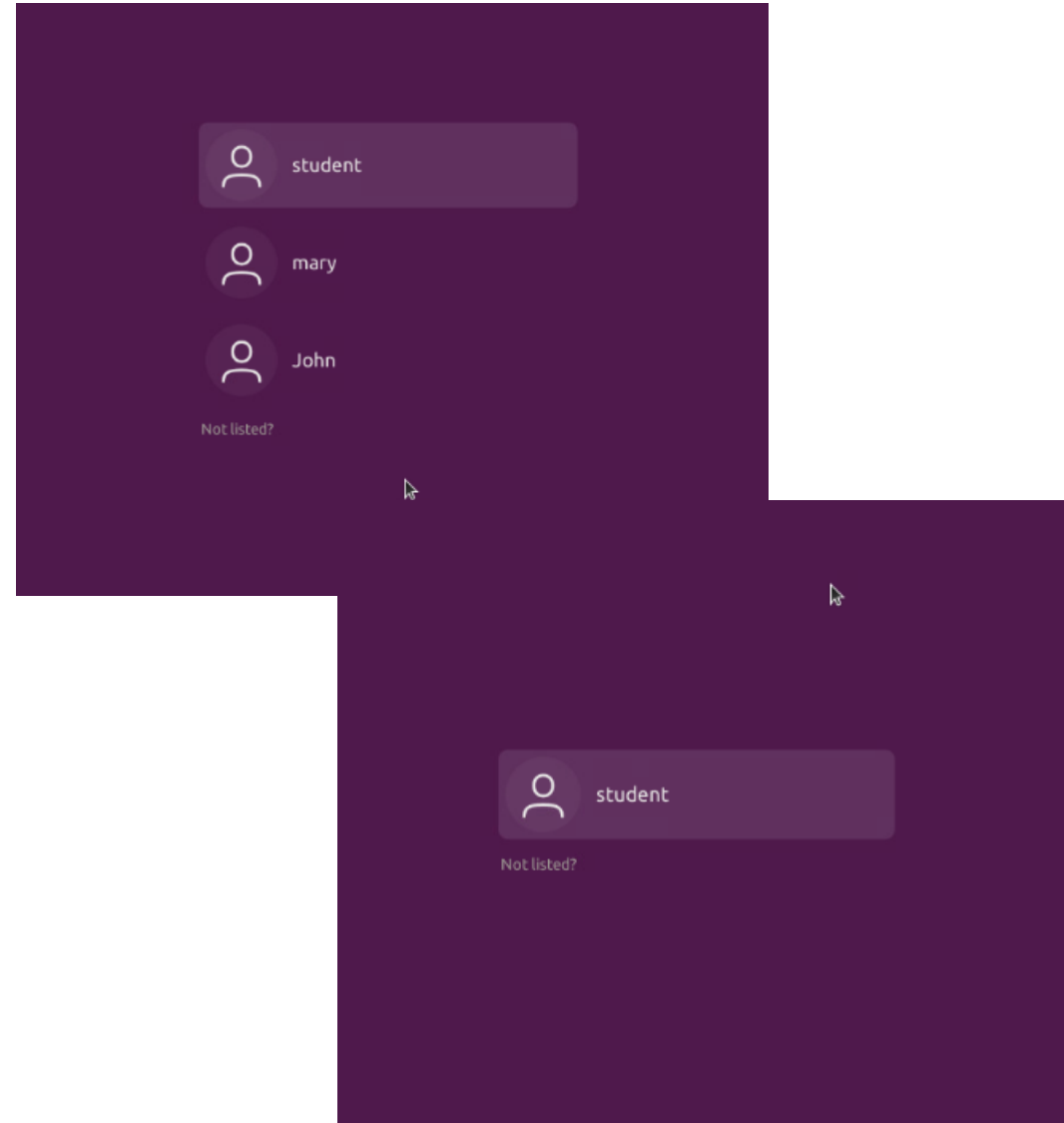
```
mary@ubuntu1: ~  
mary@ubuntu1:~$ cd  
mary@ubuntu1:~$ nano .bashrc  
mary@ubuntu1:~$ source .bashrc  
mary@ubuntu1:~$ todoist  
Enter today's to-do-list (Press ENTER to complete):  
1. school 2. school 3. school  
You entered: 1. school 2. school 3. school  
mary@ubuntu1:~$ cat MyToDoLists  
Fri 22 Mar 2024 07:04:53 PM EDT  
Today's to-do-list -- 1. school 2. school 3. school  
  
mary@ubuntu1:~$ echo "Barbara Espericueta"  
Barbara Espericueta  
mary@ubuntu1:~$ date  
Fri 22 Mar 2024 07:06:32 PM EDT  
mary@ubuntu1:~$
```


Add users in GUI



```
john@ubuntu1: ~  
john@ubuntu1:~$ cd  
john@ubuntu1:~$ pwd  
home/john  
john@ubuntu1:~$ nano .bashrc  
john@ubuntu1:~$ source .bashrc  
john@ubuntu1:~$ todolist  
Enter today's to-do-list (Press ENTER to complete):  
1. family 2. family 3. family  
You entered: 1. family 2. family 3. family  
john@ubuntu1:~$ echo "Barbara Espericueta"  
Barbara Espericueta  
john@ubuntu1:~$ date  
Fri 22 Mar 2024 07:36:55 PM EDT  
john@ubuntu1:~$
```

Add/Remove users and groups



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Operating Systems

Module 5 Network Configuration

Discovering IP configurations of the Linux machine and exploring different networking utility programs.

Discover host IP configurations

1. What is the IP address of your Ubuntu machine?

Answer here:

192.168.1.107

2. What is the IP address of its default gateway?

Answer here:

192.168.1.1

3. What is the IP address of its DHCP server?

Answer here:

192.168.1.1

4. What is the IP address of its DNS server?

Answer here:

192.168.1.1

```
nameserver 192.168.1.1
search devry.edu
student@ubuntu1:/var/lib/dhcp$ ping -c 4 192.168.1.1
PING 192.168.1.1 (192.168.1.1) 56(84) bytes of data:
64 bytes from 192.168.1.1: icmp_seq=1 ttl=64 time=0.305 ms
64 bytes from 192.168.1.1: icmp_seq=2 ttl=64 time=0.486 ms
64 bytes from 192.168.1.1: icmp_seq=3 ttl=64 time=0.320 ms
64 bytes from 192.168.1.1: icmp_seq=4 ttl=64 time=0.577 ms

--- 192.168.1.1 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3080ms
rtt min/avg/max/mdev = 0.305/0.422/0.577/0.114 ms
student@ubuntu1:/var/lib/dhcp$ echo "Barbara Espericueta"
Barbara Espericueta
student@ubuntu1:/var/lib/dhcp$ date
Sat 30 Mar 2024 12:22:40 AM EDT
student@ubuntu1:/var/lib/dhcp$
```

Manage network interfaces

1. Which DHCP message is shown in the output of the **sudo dhclient -v -r ens33** command?

Answer here:

DHCPRELEASE

2. Which four DHCP messages are shown in the output of the **sudo dhclient -v ens33** command?

Answer here:

DHCPDISCOVER

DHCPOFFER

DHCPREQUEST

DHCPACK

Use network utilities

```
student@ubuntu1:~$ sudo ifconfig eth0 up
student@ubuntu1:~$ ifconfig eth0
eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST>  mtu 1500
    inet 192.168.1.107  netmask 255.255.255.0  broadcast 192.168.1.255
    inet6 fe80::7b9e:ebf5:11a6:34e4  prefixlen 64  scopeid 0x20<link>
    ether 00:15:5d:00:ba:04  txqueuelen 1000  (Ethernet)
    RX packets 6228  bytes 487674 (487.6 KB)
    RX errors 0  dropped 0  overruns 0  frame 0
    TX packets 9717  bytes 796213 (796.2 KB)
    TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0

student@ubuntu1:~$ echo "Barbara Espericueta"
Barbara Espericueta
student@ubuntu1:~$ date
Sat 30 Mar 2024 12:52:20 AM EDT
student@ubuntu1:~$
```

Course Project

Operating Systems

Module 6 System Performance Monitoring

This module was to explore Linux utilities that monitor processes, user activities, and network usage.

Monitor Linux processes

1. What is the default action of the *15 SIGTERM* kill signal?

Answer here: Kill the highlighted process.

2. In the System Monitor window, click on *% CPU* to sort the processes by CPU load. Which process shows the highest percentage of CPU usage?

Answer here: gnome-shell (GUI)

References:

1. Project Assistance video

Monitor user activities

Issue the **sudo accton on** command to turn on GNC accounting. Run the **sudo updatedb** command. Enter **lastcomm updatedb** to check if the *updatedb* command was executed before. Remember to turn off GNC accounting (**sudo accton off**) after answering the questions.

1. What flag value is displayed in the output?

Answer here: S – “super user”

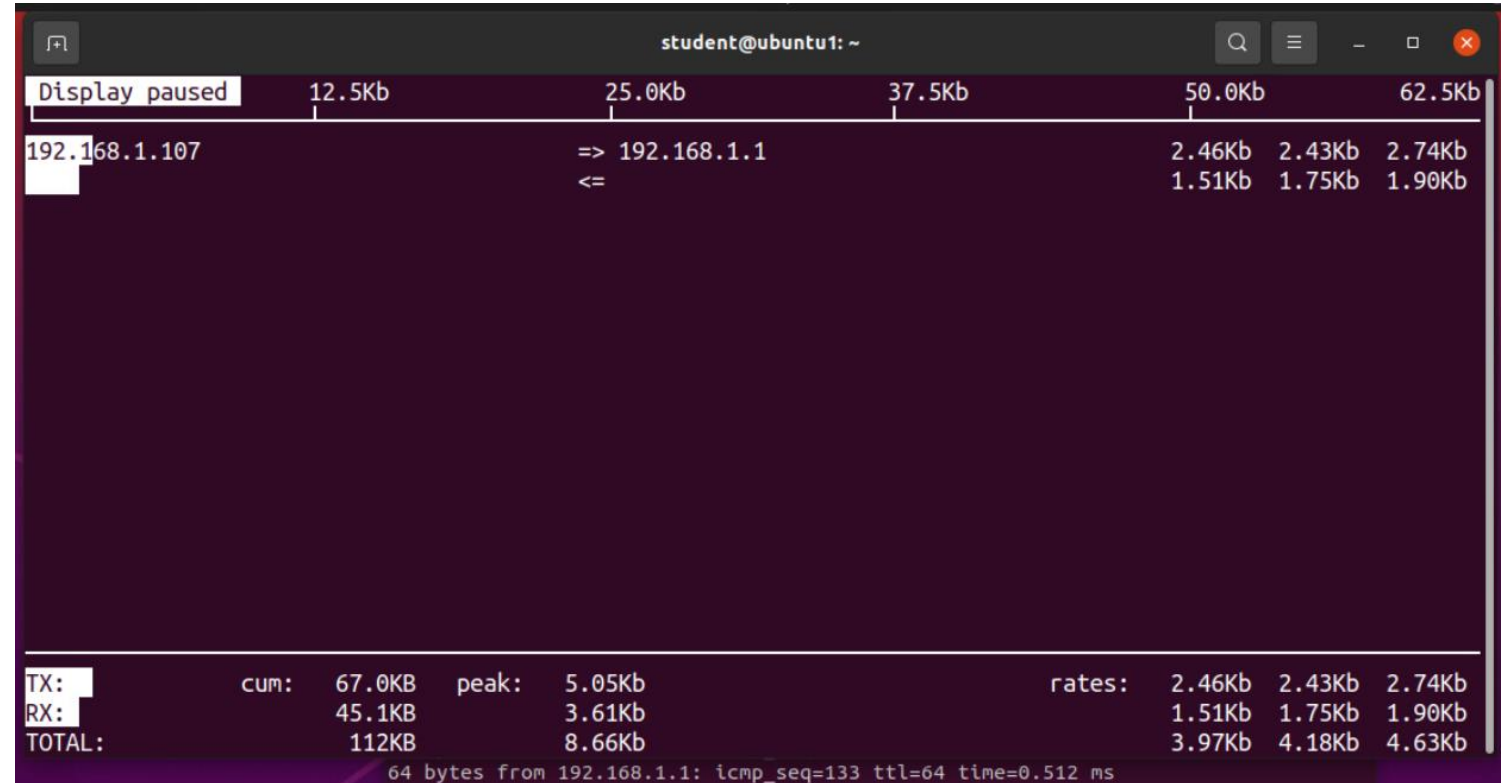
2. Why is the name of the user who ran the processes shown as root, not student?

Answer here: superuser -- since we typed sudo in front of the command.

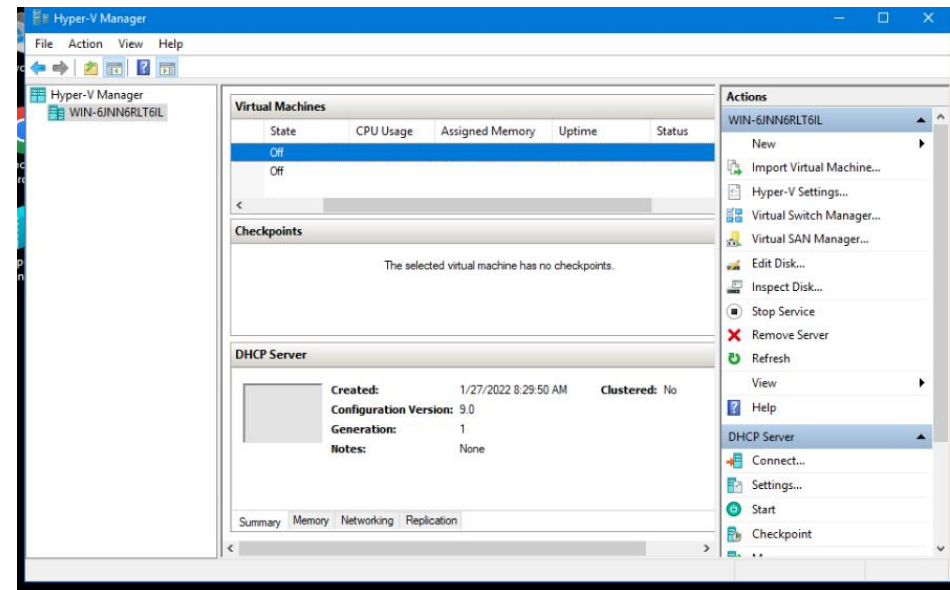
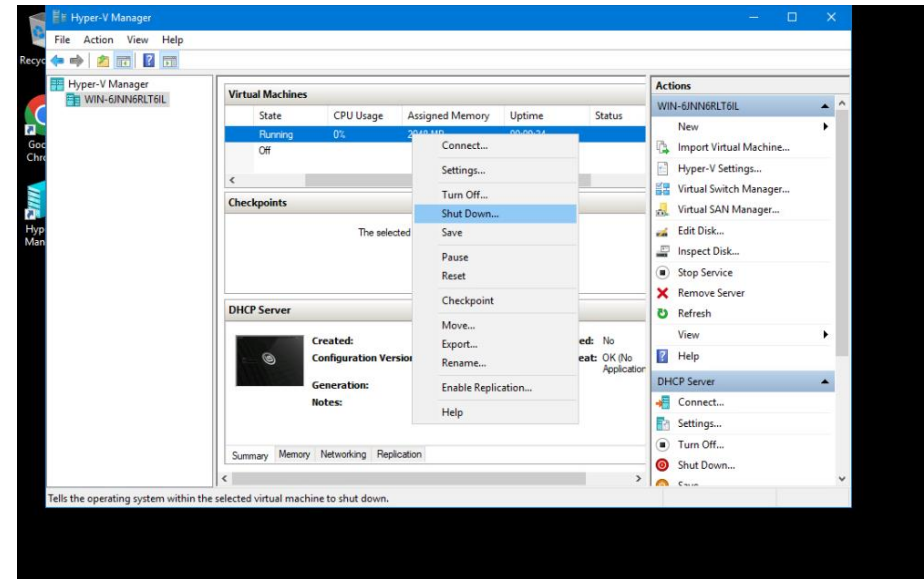
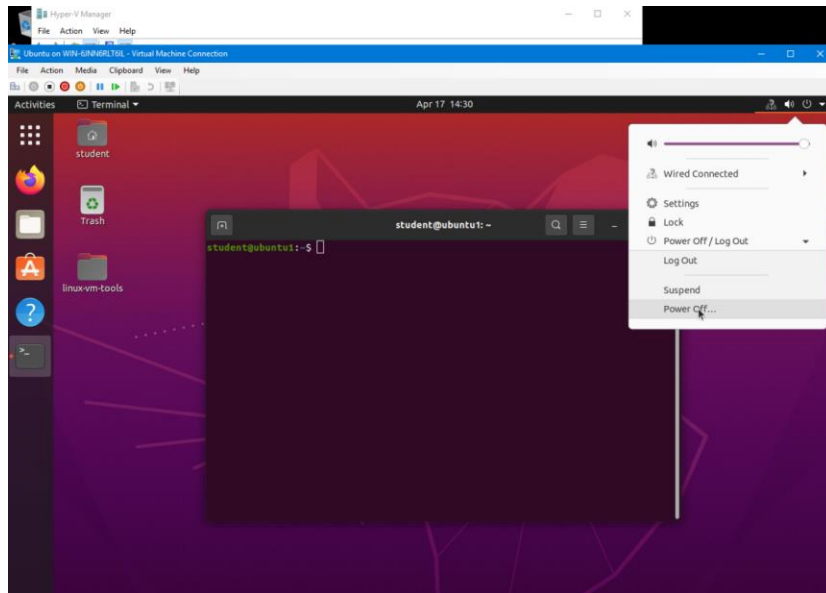
References:

1. Project Assistance video

Monitor network bandwidth usage



Properly Shutting Down Servers



Challenges/Lessons Learned

Challenges I faced during these projects were the commands, trying to memorize without looking them up repetitively. Reading my Linux+ and LPIC-1 Guidebook until I completely understood each command was how I overcame my challenge of the commands. Lessons I learned over this course were to stay persistent in the studies to deep dive into each lesson to completely understand the big picture.

Career Skills

Skills I developed that will benefit my career is obtaining Linux+ knowledge and certification, there are many opportunities with the knowledge of Linux. Adaptability, continuous learning, service, and quality focus competencies are these projects key components that will help me gain the opportunities to advance my career.

Conclusion

Learning to utilize the Linux filesystem hierarchy, shell scripts, management of users and groups, network configurations, and systems performance monitoring in the Linux operating system is crucial in Cybersecurity. Gaining competencies in adaptability and quality focus sharpens me as an individual. Having the challenges of commands and repetitive research to learn persistence, is a great trade in learning the Linux operating system.